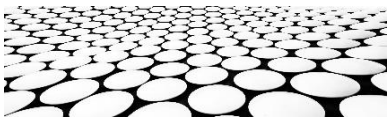
CSI4142 Fundamentals of Data Science



ASSIGNMENT 3

Predictive analysis
Regression and Classification

GOALS

The overall goal of this assignment is to become familiar with Predictive Analysis, both for regression (prediction of numerical values) and classification (prediction of a categorical value).

At the end of this assignment, you will have:

- Reviewed your Python skills, as the project MUST be done in Python
- Explored Kaggle, an amazing resource of challenges and datasets
- Programmed and evaluated a linear regression approach on a regression task
- Programmed and evaluated the impact of outlier removal on a linear regression method
- Programmed and evaluated a decision tree approach on a classification task
- Programmed and evaluated the impact of feature aggregation on both linear regression and decision trees
- Documented everything, in Jupyter Notebooks, in a way to make your analysis understandable and reproducible.

SUBMISSION INFORMATION

- Deadline:
  - Submission of your notebook: **Tuesday, March 17th, midnight**
  - Grace period (no penalty) is until 12:30… after that it's 20% per day off.
- Groups:
  - **You are expected to form groups of 2 and do a single submission per group**.
  - You need to register your group in Brightspace. As we are allowing groups to change from one assignment to another, <u>you have to register again your group.</u> This time as a A3 group.
  - It is not possible to work alone, and to enforce such policy, *a penalty of 20% will be imposed on solo groups.* Make sure you find a partner through the Group Search sheet on Brightspace. Or even use the Forum if you prefer.
- Where to submit:
  - Your submission must be done in Brightspace in Assignment section (Assignment 3)
- Submission format:
  - <u>Only Jupyter Notebook files will be accepted to show your programs</u>
  - For the datasets used in your analysis, your code MUST contain a link to the dataset so the TAS do NOT need to download any data.
- Demo/Presentation (**mandatory**):
  - You must select a time (a set of timeslots will be provided) to meet with one of the TAs to present your work.
  - Both students must be present during the presentation. If one student is missing, that student will have a 20% penalty on the entire assignment. The student who missed will not be allowed to reschedule as the demo is already done.
  - A group that has a valid reason can contact the TA to reschedule without penalty, but it must be done at least 24 hours before your time slot, and there must still be available timeslots. A group cannot reschedule twice.
  - If you miss your time slot, it is 20% penalty on the entire assignment. To not have zero, you should reschedule within 24 hours of your missed slot, and while there are still available timeslots.
  - Any no-show will be given zero on the assignment. **Even if the demo part counts for 12 marks out of 60, not doing the demo results in zero for the entire assignment.**

PLEASE NOTE:

If the corrector cannot run your code, the mark will be zero for the entire assignment. It is your responsibility to test that the code cells in your Notebook are executable. The <u>most frequent problem</u> is that the dataset is not readable (file not found). Please make sure the dataset is either read directly from a public place (a dataset repository) or read from one of your shared repository (accessible to anyone) so the TA can run your notebook without any data download.

PROGRAMMING REQUIREMENTS

1. **Choose 2 datasets**

As you know, the Kaggle site is a very good site to explore as it contains datasets for many tasks in data science and AI. There are also other sites for datasets if you wish to further explore. You must choose 2 datasets, and they cannot be the same as for your Assignment 1 nor Assignment 2.

Dataset 1 – Regression

The first dataset will serve for your empirical study of regression (described in Section 2).

For **Dataset 1**, you can choose among the 3 datasets suggested below. Or you can choose a different dataset that has a good match with a prediction task (usually Kaggle mentions the possible tasks).

Suggested Datasets

1. **Car Price Prediction Dataset**
   - **Description**: Predicts car prices based on features like mileage, brand, year, fuel type, and transmission type.
   - **Link**: Kaggle Car Price Dataset
2. **Medical Insurance Cost Prediction Dataset**
   - **Description**: Predicts medical insurance costs based on age, BMI, smoking status, region, and other features.
   - **Link**: Kaggle Medical Insurance Dataset
3. **Housing Prices Regression Dataset**
   - **Description**: Predicts house prices with a couple of binary features.
   - **Link**: Housing Prices Regression

Dataset 2 – Classification

The second dataset will serve for your empirical study of classification (described in Section 3).

For **Dataset 2**, you must select a dataset that is different from Dataset 1. Also the suggested datasets for Dataset 1 are for regression analysis and you must find a dataset for classification analysis, so they would not qualify.

Furthermore, your second dataset CANNOT be a synthetic dataset and it must be in a domain different from the first dataset. I want you to explore 2 different domains. Another constraint is to find a dataset with a minimum of 5 columns so you have various features to explore for the classification.

2. Regression Empirical Study with Linear Regression

In this part of the assignment, you must perform an empirical study in which you evaluate a linear regression approach on a regression task.

a) Cleaning the data
   a. You have programmed in Assignment 2 some validity checks and imputation methods. Reuse your methods here to clean your data.
b) Categorical feature encoding
   a. Linear regression applies to numerical features. We have seen in class that one-hot encoding can transform categorical features into numerical features. Apply such encoding.
c) EDA and Outlier detection
   a. We explored in class different approaches for outlier detection, one of them being Local Outlier Factor (LOF). Program such approach for outlier detection.
   b. As LOF is costly to use (if large dataset), first use EDA to visualize the data and find which feature LOF should be used on. If there are no features with outliers in your dataset, then purposely introduce some outliers for at least one feature so you can use LOF to detect them.
   c. Decide what to do with outliers. Either remove them, or consider them as missing values on which to apply imputation methods.
d) Predictive analysis: Linear regression
   a. Explore the *LinearRegression* method suggested in scikit-learn (or other packages). scikit-learn also contains Ridge, Lasso and ElasticNet regressions if you want to explore those (optional).
e) Feature Engineering
   a. Program a feature aggregator to create 2 additional features. We will explore aggregation during March 11th lecture, but you can go ahead earlier and program something simple (e.g. adding two features). Think of interesting ways to combine the features based on the dataset and what could make sense. The purpose of aggregation is to create new features from the ones you have.
f) Empirical study
   a. Split your data into 3 subsets for train, validation and test sets.**
   b. Decide on some evaluation metric(s) (MSE, LMSE, R2). Choose at least one.
   c. As baseline, use the linear regression with baseline settings, without outlier removal and without feature aggregation. Using your chosen metric(s), evaluate on the validation set using a 4-fold cross-validation.
   d. Try different (at least 3) system configurations (e.g. with/without dealing with outliers, with/without feature aggregation). Using your chosen metric(s), evaluate the different system configurations on the validation set using a 4-fold cross-validation. Based on those experiments, you will choose the best configuration.
   e. Retrain your best configuration on train and validation combined and use the Test Set to perform your final evaluation. This is to simulate a future test at a client's site. The Test Set should NOT have been used before this final evaluation.

g) Result analysis
   a. Analyse the obtained results and show if any improvement was achieved with different settings.
   b. Discuss the impact of outlier detection and feature aggregation.
   c. Discuss how the results on the unseen test set compare to the cross-validation results.

3. Classification Empirical Study with Decision Trees

In this part of the assignment, you must perform an empirical study in which you evaluate a decision tree approach on a classification task.

a) Cleaning the data
   a. (same as for regression study)
b) Numerical feature encoding (optional)
   a. Decision trees might benefit from binning of numerical values. You can explore this idea, but it's not mandatory.
c) EDA and Outlier detection
   a. (same as for regression study)
d) Predictive analysis: Decision Trees
   a. Explore the *DecisionTreeClassifier* method suggested in scikit-learn (or other packages).
   b. Look at the parameters (splitting criterion (gini), max_depth, min_samples_split, etc) and choose a baseline setting.
e) Feature Engineering
   a. (same as for regression study)
f) Empirical study
   a. Split your data into 3 subsets for train, validation and test sets.**
   b. Decide on some evaluation metric(s) (Precision, Recall, F1, Accuracy). Choose at least one.
   c. As baseline, use the *DecisionTreeClassifier* method with baseline settings, without outlier removal and without feature aggregation. Using your chosen metric, evaluate on the validation set using a 4-fold cross-validation.
   d. Try different (at least 3) system configurations (e.g. with/without dealing with outliers, with/without feature aggregation). Also try different parameter settings for the *DecisionTreeClassifier* method (changing the max_depth, min_samples_split). Using your chosen metric(s), evaluate the different system configurations on the validation set using a 4-fold cross-validation. Based on those experiments, you will choose the best configuration.
   e. Retrain your best configuration on train and validation combined and use the Test Set to perform your final evaluation. This is to simulate a future test at a client's site. The Test Set should NOT have been used before this final evaluation.

g) Result analysis
   a. Analyse the obtained results and show if any improvement was achieved with different settings.
   b. Discuss the impact of outlier detection and feature aggregation.
   c. Discuss how the results on the unseen test set compare to the cross-validation results.

**** PLEASE NOTE:** In class, I mentioned that we need to split our training data into train/validation. Here I'm asking you to split into train/validation/test. So, first, remove a test part (e.g. 15%) of your dataset. This is to simulate your client's data (if you were a consultant doing a study) that they would gather in the future, once your system is in production. It would never be accessible to your system. So you remove it and never show it to our algorithm during training. What is left (original train less test), let's call it train_reduced. Then split train_reduced into train/validation to do the cross-validation and finetune your system.

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EVALUATION (60 points)

Jupyter Notebook (48 points)

- Overall effort in the reports (4 points)
  - Provided all required steps clearly identified with different sections
  - Good cell separation (text, code, results, etc)
  - Tests on various examples easy to perform by the corrector
  - Report detailed enough for reproducibility
- Heading (-3 if missing)
  - Group number + student names + student numbers
  - Work split (provide information about who did what for the assignment, explain the work split between the 2 students as to know who was responsible for what)
- Datasets descriptions (4 points)
  - Dataset name, author, purpose (what was it made for)
  - Shape: how many rows and columns
  - A list of the features + descriptions (what do they represent and are they categorical or numerical)
- Data Cleaning, Outlier Detection, feature aggregation (10 points)
  - Data cleaning tools used on both datasets
  - Programming of outlier Detection with LOF and decision on how to deal with outliers for both datasets
  - Programming of feature aggregation to create at least 2 additional features on both datasets
- Linear Regression study (15 points)
  - Correct use of train/validation/test
  - Correct use of metrics
  - Correct experiments (and at least 3 configurations)
  - Clear presentation of results (comparative table + analysis)
- Classification study (15 points)
  - Correct use of train/validation/test
  - Correct use of metrics
  - Correct experiments (at least 3 configurations)
  - Clear presentation of results (comparative table + analysis)
- References (-3 if missing)
  - For any part of your code taken from a web site (even a tutorial site or stackoverflow), you must provide the reference to it.
  - If part of your code was generated using Generative AI, please list the tool as well as the queries performed for each code snippet used.
  - Any theory/algorithms found in books, slides, tutorials that you used should be referenced.

Presentation/Demo (12 points)

- Understanding of the work
  - Students show a good understanding of the work they have done.
- Clarity of explanations
  - Students clearly express themselves using appropriate terms to describe what they have done.

QUESTIONS

- You can ask your questions within the assignment topic of the discussion forum on Brightspace.
- To make sure you get answers on the forum on a regular basis, the 3 TAs will share the days to answer your question. Notice that they are NOT available on weekends.

	Forum Monitoring						
	Monday	**Tuesday**	**Wednesday**	**Thursday**	**Friday**	**Saturday**	**Sunday**
Morning (before noon)	Svetlana	Ángel		Ángel	Svetlana		
Evening (before 9pm)	Shavbo	Shavbo	Svetlana	Shavbo	Ángel		

If you need to reach the TAs by email, you should use the following emails:

- Shavbo: ssale038@uottawa.ca
- Svetlana: sesina@uottawa.ca
- Ángel : avale073@uottawa.ca